

Bike Rental and Ride-Sharing Platform



Module: SE1020 - Object Oriented Programming

Technology Stack: Java, Spring Boot, Thymeleaf, Bootstrap, File Handling

Repository: <https://github.com/ashifahmed-924/OOP-bike-rental-system>

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Date: 2026-04-23

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Introduction

This project is a web-based Bike Rental and Ride-Sharing Platform developed for the SE1020 Object Oriented Programming module. The system allows users to register, log in, manage bikes, create ride bookings, and handle ride-sharing requests. The backend is implemented using Java and Spring Boot, while file handling is used for data persistence instead of a database.

Objectives

- Apply object-oriented programming concepts such as encapsulation, inheritance, abstraction, and polymorphism.
- Implement CRUD operations for user management, bike management, and ride bookings.
- Develop a user-friendly interface for different system roles.
- Use file read/write operations for persistent storage.
- Track development using Git and GitHub.

Technologies Used

Backend

- Java
- Spring Boot
- Spring MVC
- Thymeleaf

Frontend and Tooling

- HTML/ CSS
- Bootstrap
- Maven
- GitHub

Main Features

- User registration and login with role-based access.
- Admin user management with create, search, update, and delete operations.
- Operator bike management with add, update, and delete operations.
- Rider bike booking and ride-sharing request handling
- Operator ride booking overview limited to bikes added by that operator.
- File-based persistence using text files.

Project Structure

- `src/main/java/app`- Spring Boot application and controllers
- `src/main/java/model`- domain models and polymorphic user types
- `src/main/java/dao`- file-based data access objects
- `src/main/resources/templates`- Thymeleaf HTML templates
- `src/main/resources/static/assets`- CSS and image assets
- `src/main/webapp/data` - text files for persistence

Object-Oriented Programming Concepts

Encapsulation

Encapsulation is used in the model classes through controlled access to data fields using getters and setters. Examples include Bike, RideBooking, and ShareRideRequest.

Inheritance

Inheritance is used through the class hierarchy Person -> User -> AdminUser / OperatorUser / RiderUser. This structure promotes reuse of common properties such as username and password.

Polymorphism

Polymorphism is implemented using specialized user subclasses. The system works with the abstract `UserType`, while each subclass provides its own implementation of `getRole()` and `getDashboardPath()`. This allows the application to redirect users based on behavior defined by the actual runtime object.

Abstraction

The classes `Person` and `User` are abstract and define the common structure and required methods for all user types.

CRUD Operations

Module	Create	Read	Update	Delete
Users	Register / Add User	Search / List Users	Update User	Delete User
Bikes	Add Bike	List Bikes	Update Bike	Delete Bike
Ride Bookings	Create Booking	List Bookings	Update Booking	Delete Booking

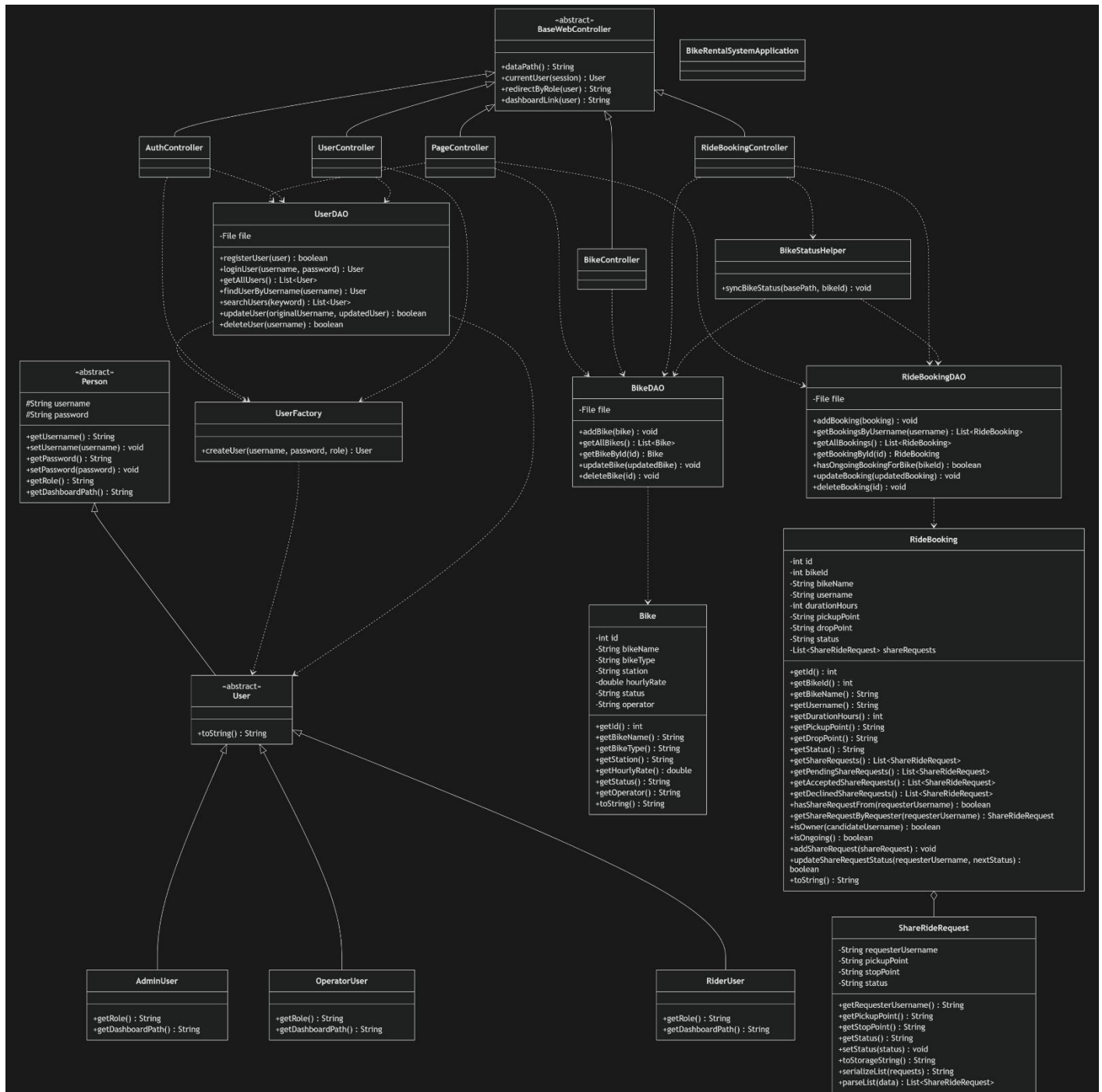
File Handling

The application stores data using text files instead of a relational database.

- `users.txt` - stores usernames, passwords, and roles
- `bikes.txt` - stores bike records
- `ride-bookings.txt` - stores ride booking and share request details

File operations are handled through `UserDAO`, `BikeDAO`, and `RideBookingDAO`.

Class Diagram



High Quality Image: <https://drive.google.com/file/d/1-YZAgOlzR-D6BbWroAVAx4r8OSSiC8a5/view?usp=sharing>

User Interfaces

- Home page
- Login and registration pages
- Admin user management page
- Operator bike management page
- Ride booking management page

Challenges Faced

- Migrating the project from JSP/Servlet architecture to Spring Boot.
- Converting JSP pages into Thymeleaf templates.
- Preserving file-based data storage after the framework migration.
- Adding a clearer polymorphism implementation for viva evaluation.

Git Repository and Commit History

Repository URL: <https://github.com/ashifahmed-924/OOP-bike-rental-system>

Version control for the project was managed using Git and GitHub. The commit history shows the progress of the application from the initial implementation to the final Spring Boot and Thymeleaf migration.

Commit ID	Message
681ffa4	Convert app to Spring Boot and Thymeleaf
10d8d84	Initial Project structure